

Serial No.	Questions
1	Define Group.
2	Define Sub-Group of a Group.
3	Define order of Group and Order of an element of Group.
4	Let $C_1 = (5\ 2\ 3)$, and $C_2 = (1\ 4\ 6\ 7)$ be cycles in S_8 . Find $C_1 C_2$ and express it as product of transpositions.
5	Give an example of a monoid which is not a group.
6	Let $a, b \in (G, *)$, where G is a Group. Show that $(a * b)^{-1} = b^{-1} * a^{-1}$.
7	Define i) a permutation, ii) a symmetric group.
8	Let $\alpha = (1\ 3\ 2\ 5)(1\ 4\ 3)(2\ 5) \in S_5$, Find α^{-1} and express it as a product of disjoint cycles. State whether $\alpha^{-1} \in S_5$.
9	Show that in any Group $(G, *)$, Identity and inverse of any element is unique.
10	Show that in a Ring R with identity element 0 and unity element 1, $\forall a \in R, a \cdot 0 = 0 = 0 \cdot a$.
11	Prove that $F = \{a + b\sqrt{2} \mid a, b \text{ integers}\}$ is an integral domain but not a field.
12	In a group $G = (A, *)$. Show that if $a^2 = e$ for all a in a group G , then G is commutative.
13	Let $(R, +, \cdot)$ be a Ring with unity. Show that $\forall a, b \in R$, 1. $(-1) \cdot a = -a$ 2. $(-1) \cdot (-1) = 1$ 3. $(-a)(-b) = ab$
14	Let Z_n denote the set of integers $\{0, 1, 2, \dots, n-1\}$. Let $*$ be binary operation on Z_n such that $a * b = \text{the remainder of } a + b \text{ divided by } n$ Construct the table for the operation $*$ for $n=5$ and using it show that $(Z_5, *)$ is a group.
15	Consider the set $\mathbb{Q}$ of rational numbers, and let $*$ be the operation on $\mathbb{Q}$ defined by $a * b = a + b - ab$ (a) Find $4 * 7$, $9 * (-3)$ and $6 * 2/5$.

	(b) Is $(\mathbb{Q},*)$ a semigroup? Is it commutative? (c) Find the identity element for $*$.
16	Consider the Group $G = \{1,2,3,4\}$ under multiplication modulo 5. (a) Find the multiplication table of G. (b) Find $2^{-1}, 3^{-1}, 4^{-1}$.
17	Show that $Z_7 = \{0,1,2,3,4,5,6\}$ is a Ring under addition and multiplication modulo 7.
18	Consider the ring $Z_8 = \{0,1,2,\dots,7\}$ of integers modulo 8. (a) Find the units of Z_8. (b) Find $-3, -7$, and 3^{-1}.
19	Show that $R = \{0,2,4,6,8\}$ is an integral Domain under addition and multiplication modulo 10.
20	Show that set of fourth roots of unity is a Group under multiplication.
21	Define ring with an example.
22	Define Integral domain with an example.
23	Define Field with an example.
24	Show that Z_{11} is a field under addition and multiplication modulo 11.